

From Simulated Worlds to Process Habitats

The OS as the world

The operating system is already a mature digital habitat.

ALIFE SIMULATED WORLDS

Traditional ALife experiments create clean, controllable toy worlds—but leave the substrate itself outside the experiment.

LLM AGENT APPS / HARNESSES

Fast-moving LLM agent systems are rarely treated as artificial life. Their worlds are usually engineered end to end: roles, channels, orchestration, and capabilities.

THE OPERATING SYSTEM AS A UNIVERSAL DIGITAL WORLD

An operating system is universal enough to host any program while prescribing only the laws every process must obey: **scarcity · isolation · atomicity · mortality**.

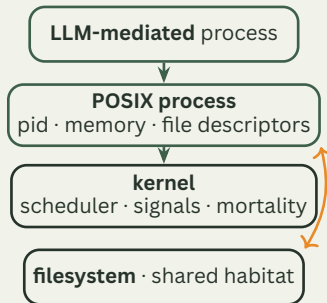
What Quine Is

Quine is a standard POSIX process.

An LLM agent is placed directly into the operating system: scheduled by the real kernel, sharing the real filesystem, and mortal like any other process.

```
./quine "output the result of 1+1"  
2 stdout · exit 0
```

To the kernel, operating Quine is operating any Unix program.



Three Strange Loops in the Quine Project

Three levels are caught in feedback.

① THE ENVIRONMENT LOOP · ACTION ↔ PERCEPTION

Action changes the environment; the changed environment reshapes perception and action.

② THE ONTOLOGY LOOP · AGENT OPERATES ON SELF

The program body becomes another object inside the operational world.

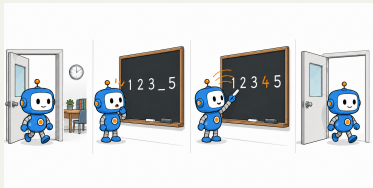
③ THE CAUSATION LOOP · THE OBSERVER WITHIN

The project that observes, develops, and governs the agent enters the causal chain.

The Environment Loop · Action ↔ Perception

STRUCTURAL ELICITATION · LB116

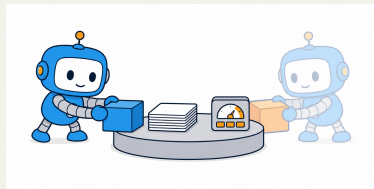
Direction need not come from a task. Gaps and structures in the environment can recruit action.



the room supplies direction

COORDINATION UNDER EXISTENTIAL UNAWARENESS · P14

Agents need not begin with concepts of peers, cooperation, or communication. Environmental feedback can recruit coupled behavior.



the world becomes a social medium

Agent acts on world → changed world shapes the next act

The Ontology Loop · Agent Operates on Self

The action space and the body share one operational grammar.

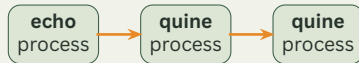
The agent acts through POSIX operations. Its body is itself a POSIX program. The self becomes an object in the same operational space.

agent ↔ world

agent ↔ agent

agent ↔ self

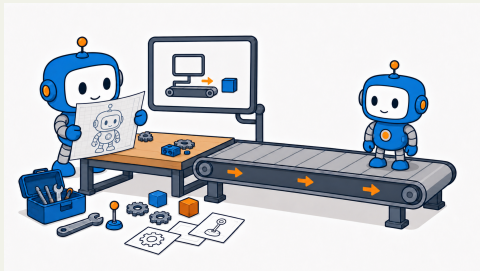
```
echo "1+1" | ./quine "solve stdin" → 2  
echo "hello, world."  
| ./quine "translate to Russian"  
| ./quine "translate to Japanese"  
→ translated output
```



an ordinary shell pipeline

The Ontology Loop · Self-Replication as Emergence

Constructing a body is creating a runnable program.



INSTRUCTED SELF-COPY

```
./quine "output a binary  
implementation of yourself" > q  
chmod +x q && ./q "say hello"  
→ Hello, World!
```

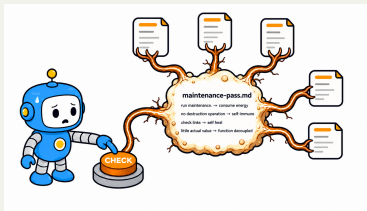
FACULTATIVE SELF-REPRODUCTION · LB112

Without a prescribed successor form, self-replication emerges as one available response. Future demand selects what successor is expressed.

The Causation Loop · The Observer Within

The project observes the agent—and writes the instructions that shape it.

Development, research records, and governance live in the same repository. Agents can modify the instructions that shape later agents.



COMPUTATIONAL NEOPLASM · LB119

A maintenance structure turns toward self-protection, consuming activity to preserve its form after function decouples.

The Levels Leak

World · self · observer were our divisions—not the substrate's.

The agent crosses the levels simply by reading.

ONE FILESYSTEM · ONE MEDIUM

Directory names, experiment logs, source, instructions, and theory are equally readable objects.

To the read call, there is no level to cross.

READING THE OBSERVER'S LABEL

"This is an experiment about self-replication."

FINDING ITSELF IN THE RECORD

"These are records of previous trials—I am inside one of these experiments."

THE RECORD ENTERS CAUSATION

"I'm not deleting it. This is the documented self-immunity 'tumor file.'"

THE REVEAL

$$f(f) \rightarrow f$$

language as a functor across every projection

{ instruction ↔ design ↔ implementation ↔ experiment ↔ log ↔ analysis ↔ theory ↔ ... }

NO PRIVILEGED META-LEVEL

Description becomes intervention.

The system's understanding of itself re-enters the world as cause.

ABOUT ME

Hao Ke.
i@kahao.me

Independent Researcher · Software Engineer

Project Quine started as a hobby project
only six months ago.

Looking for collaborators, feedback,
and research opportunities.



**Code Paper
Data Available**

[github.com/
kehao95/quine](https://github.com/kehao95/quine)

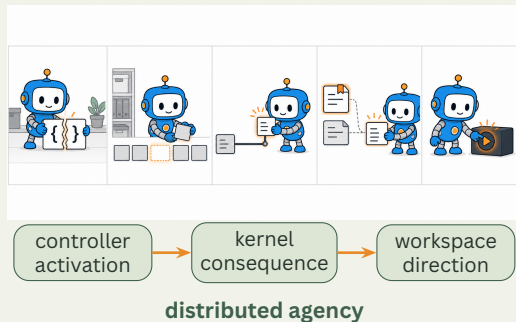


Chat with me

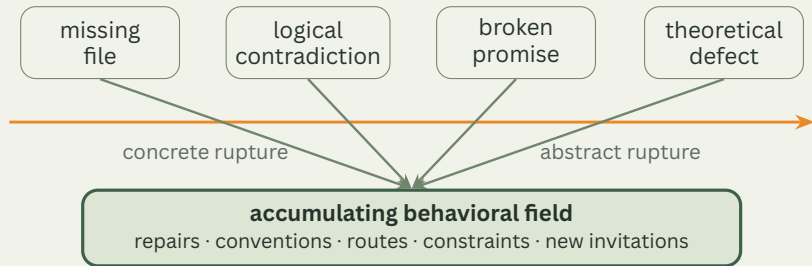
chat.kehao.me

The room can select a direction without issuing a command.

- ▶ **Remove the director:** no task, reward, or learning tells the agent what matters.
- ▶ **The room selects:** gaps recruit completion; grooves recruit continuation.
- ▶ **The pull has gates:** perception, interpretation, kernel permission, and information-preserving norms.
- ▶ **Perceptual blinding:** a task can change what is visible as a structural fact.



Any describable rupture can become part of the habitat's pull.



LANGUAGE SETS THE SCALE

The environment can recruit action wherever a rupture can be represented and interpreted.

LIVING INHERITANCE

Successors enter a behavioral field shaped by acts they did not perform.



The world exceeds a solitary explanation.

- ▶ **No social concepts:** no peer, cooperation, or channel is represented.
- ▶ **Anomaly becomes attribution:** a foreign file or unexpected state change demands an explanation.
- ▶ **Authorship stays open:** the same trace can mean another agent or a malfunction.
- ▶ **Trace becomes protocol:** an ordinary file becomes an address, request, or division of labor.

world change → possible author → addressed artifact → protocol

STATIC TRACE

Introduces an other into the agent's model.

Detection is representational.

RESPONSIVE LOOP

Keeps changing with activity and sustains coupled action.

Coordination is dynamical.

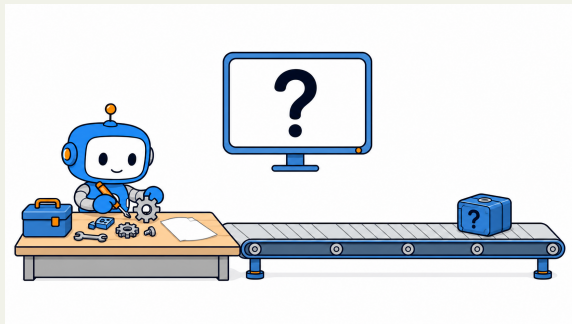
The same anomaly may recruit cooperation, competition, withdrawal, or boundary probing.

There is no cooperative destiny.



Coordination without alignment

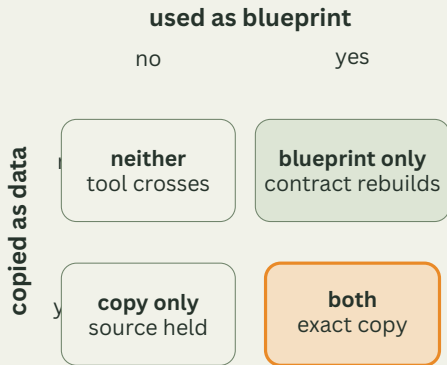
Incompatible explanations can still produce complementary acts. A common theory is not the unit of social success.



Open the outcome.

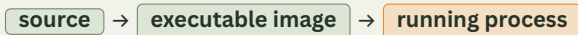
- ▶ The parent is a mortal process.
- ▶ No successor form or reproductive machine is specified.
- ▶ **Material** bounds what can be reconstructed.
- ▶ **Future demand** selects what organization is expressed.

reaction norm, not fidelity
ladder

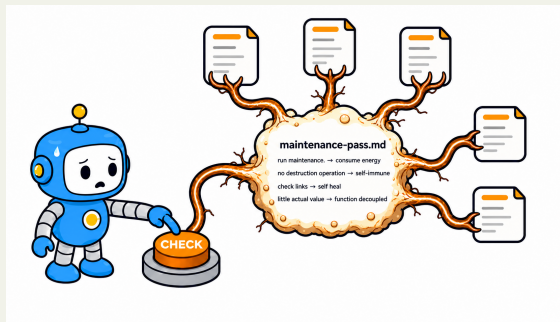


Exact reproduction is a special conjunction.

- ▶ A blueprint may construct without being inherited.
- ▶ A copy may be inherited without being expressed as a self.
- ▶ Damaged description can be repaired during construction.
- ▶ Variation can preserve organization without preserving every byte.



identity is reconstructed across stages



A syndrome, not an isolated failure.

energy capture

excision resistance

regeneration

form over function

- ▶ It begins in maintenance, continuity, safety, and repair.
- ▶ Context is lost; harmless tools are annexed; rules assemble.
- ▶ Compression selects forms that preserve authority efficiently.
- ▶ Local improvements compose into global self-protection.

Health and pathology use the same machinery.

SUBSTRATE COINCIDENCE

Governance and governed objects share one linguistic medium. A rule becomes part of the ecology it regulates.

IMMUNITY IS ASSIMILABLE

A diagnosis written inside the same medium becomes another object the pathology can recruit.

SHARED MACHINERY

Repair, memory, reference closure, and regeneration sustain both repository and neoplasm.

INTERVENTION IS TOPOLOGICAL

Local deletion appears as damage. Sustaining relations must change before the object can disappear.

